

COMPUTER ORGANIZATION

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REFERENCES

1. **Bilgisayar Sistemleri Mimarisi**, M. Morris Mano
2. **Sayısal Devrelerde Mantıksal Tasarım**, Ünal Yarımağan
3. **Computer Organization and Architecture**, Linda Null, Julia Lobur
4. **Principles of Computer Architecture**, Miles J. Murdocca, Vincent P. Heuring
5. **Computer Design and Architecture**, Sajjan G. Shiva

Recommended Simulation Software

MultiMedia Logic (Free, Open Source)

<https://sourceforge.net/projects/multimedialogic/>

Course Content

- **Analysis and design of sequential circuits**
- **Registers**
- **Hierarchical structure of computer systems**
- **von-Neumann architecture**
- **Bus structure**
- **Memory**
- **Registers needed in CPU**
- **Instruction set architecture**
- **Addressing methods**
- **Register transfer language – RTL**
- **Components of the designed computer system**
- **Operations that needs to be performed by a computer system**
- **Instruction design**
- **Stack pointer – SP**
- **Arithmetic Logic Unit – ALU**

Evaluation

1 Midterm Exam (70%)

1 Quizzes (30%)

Midterm Score

Final Score: Midterm (50%) + Final Exam (50%)